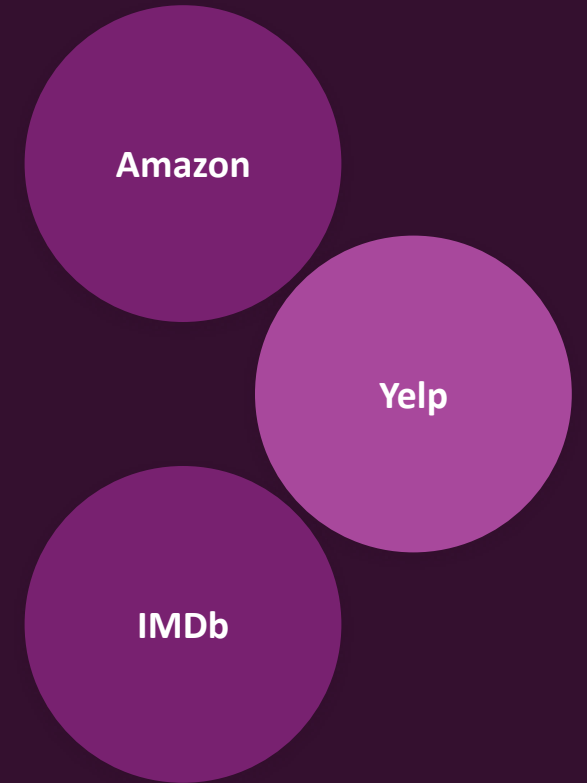


From Raw Reviews to Ready Business Actions

Businesses collect thousands of unstructured customer reviews, but turning them into clear, evidence-backed decisions takes days of manual reading. This project designs a GenAI-powered analysis pipeline that reads raw review text and automatically outputs themes, root causes, and prioritized recommendations — no manual tagging required.



Prepared for: Head of Customer Experience

Dataset Overview & Scope

Three review platforms, one unified analysis pipeline



Amazon

Product reviews

1,000 reviews



Yelp

Restaurant reviews

1,000 reviews



IMDb

Movie reviews

1,000 reviews

3,000

Total reviews analyzed

3

Independent platforms

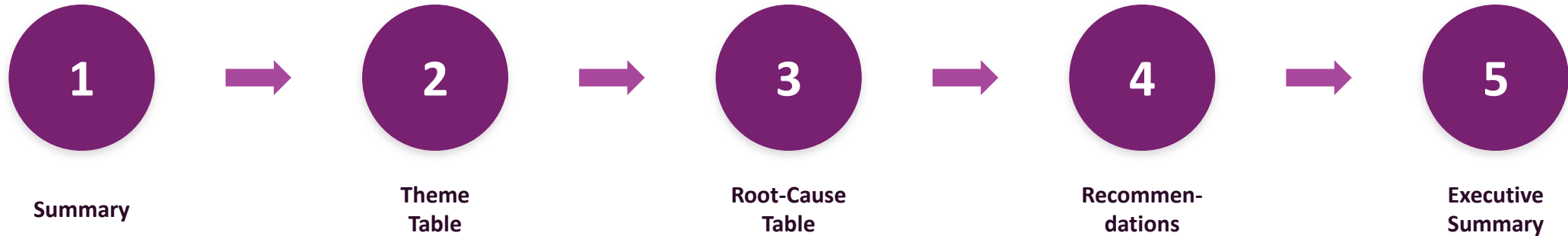
50 / 50

Positive / negative split, every dataset

Each row contains a raw review sentence plus a binary sentiment label (1 = positive, 0 = negative). Labels are perfectly balanced across all three sources, giving a clean baseline for theme and sentiment analysis.

GenAI Workflow: The Master Chaining Prompt

One assistant prompt, five chained outputs — designed to run on any new review dataset



PROMPT DESIGN — SYSTEM ROLE EXCERPT

“You are the Customer Feedback Insights Assistant. You will receive a fresh set of customer review sentences (with or without sentiment labels) from any platform. Execute the full analysis pipeline automatically and return all five outputs in sequence: Summary → Theme Table → Root-Cause Table → Recommendations Table → Executive Summary.”

Platform-agnostic — validated on Amazon, Yelp, and IMDb

Prompt Strategy: 3 of the 6 Chained Prompts

PROMPT 2

Theme Identification

Clusters reviews into named themes with sentiment, evidence, business meaning, and frequency.

PROMPT 3

Root-Cause Analysis

Separates stated evidence from inferred assumptions and assigns a confidence rating to each cause.

PROMPT 4

Recommendations

Converts each theme into one prioritized, actionable business intervention with expected impact.

REAL INPUT → OUTPUT EXAMPLE (AMAZON, LABEL = 0)

“Tied to charger for conversations lasting more than 45 minutes. MAJOR PROBLEMS!!”

Theme (Prompt 2):	Battery & Charging Problems — Negative
Root Cause (Prompt 3):	Sub-standard battery components — Confidence: Medium
Action (Prompt 4):	Audit battery suppliers; add charge-cycle QA testing — Priority: High

Key Findings: Merged Theme Table (Amazon + Yelp)

Highest-frequency themes across 2,000 product and restaurant reviews

Theme	Sentiment	Platform	Frequency
Service & Staff	Positive	Yelp	~76 reviews
Comfort & Fit	Positive	Amazon	~70 reviews
Food Taste & Quality	Positive	Yelp	~52 reviews
Value for Money	Positive	Amazon	~46 reviews
Service & Staff Problems	Negative	Yelp	~63 reviews
Comfort / Fit Issues	Negative	Amazon	~61 reviews
Won't Return	Negative	Yelp	~56 reviews
Battery & Charging Problems	Negative	Amazon	~48 reviews

TAKEAWAY

Comfort/fit and service consistency are the two most-repeated themes overall — one product-side (Amazon), one people-side (Yelp) — each appearing on both the positive and negative side of its dataset.

Recommended Business Actions

Highest-priority interventions, ranked by frequency and severity

Theme	Recommended Action	Priority
Comfort / Fit Issues (Amazon)	Run ergonomic user-testing and redesign ear-clip fit and materials.	High
Battery & Charging (Amazon)	Audit battery suppliers; add mandatory charge-cycle QA testing.	High
Service & Staff Problems (Yelp)	Standardize service training with periodic mystery-shopper checks.	High
Won't Return (Yelp)	Launch service-recovery outreach for guests reporting a bad visit.	High
Long Wait Times (Yelp)	Review peak-hour staffing; add a food-runner role during rushes.	High
Product Malfunction (Amazon)	Strengthen early-life QA inspection before packaging/shipping.	High

TAKEAWAY

These six actions were rated High because they combine the largest review volume with the most direct impact on core product or service function. Medium/Low-priority actions (value framing, menu variety, connectivity) are addressed in the full recommendations table.

Risks, Limitations & Validation Notes

1

No explicit sarcasm or ambiguity detection yet

Binary sentiment labels can't capture irony or mixed reviews (e.g., “great, another battery that dies in a day”). This is a known gap in the current pipeline, not yet a built-in step.

2

Root causes are inferred, not confirmed

Most root-cause confidence ratings are Low or Medium — review text alone can't verify operational causes like staffing or supplier issues; internal data would be needed to confirm.

3

Keyword-based theme counting can double-count

A single review mentioning two issues (e.g., “slow AND rude service”) is counted under both themes, so frequency estimates are directional, not exact.

4

Recommended validation step

Manually spot-check a 5–10% sample of borderline or ambiguous reviews before finalizing any business action derived from this analysis.

Next Steps & Improvements



Extended to IMDb — already piloted

The same five-step pipeline was run on IMDb movie reviews with zero prompt changes, proving the design is platform-agnostic.



Automate batching

Schedule the pipeline to run automatically on new review exports on a recurring basis.



Add ambiguity / sarcasm detection

Build an explicit pre-processing step to flag ironic or mixed-sentiment reviews before theme extraction.



Build a lightweight front-end

An Excel macro or simple app so non-technical stakeholders can run the pipeline on demand, without a chat interface.

Goal: move from one-time analysis to a repeatable, self-serve insights pipeline.